DMET 909 • VIDEO GAME DESIGN

FINAL APPROACH

Assignment 02 — Game World, Mechanics & Prototype

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Course: DMET 909 — Video Game Design

Project version: Assignment 02 prototype

Submission links

Deliverable	Editable link placeholder
GitHub repository	https://github.com/Darsh2984/FinalApproach-VGD-A01-
Gameplay demonstration (Google Drive)	https://drive.google.com/file/d/1SmopwkuLNCGuKtm9kaKBW1Skbz-q0nOG/view?usp=sharing

1. Executive overview and concept

1.1 Working title

Final Approach

1.2 Concept statement

Final Approach is a single-player, physics-based arcade javelin game in which rhythmic run-up input, launch angle, release position, and stability combine to determine a measurable throw. The player prepares in an interactive locker room before entering a stadium for a three-attempt precision challenge.

1.3 Genre and differentiators

Primary genre: arcade sports / precision sports. Secondary design characteristics: timing, physics simulation, short-session score chasing, environmental interaction. The distinctive feature is the transparent relationship between preparation, speed, release timing, and projectile flight rather than a single-button distance roll.

1.4 Audience and platform

Intended audience: approximately ages 13–35, especially players interested in athletics, score chasing, rhythm-based input and rapid replay. This audience and the PC target are inherited from the earlier Assignment 01 concept; they are design intentions rather than measured demographic findings. Current development target: Windows PC using Unity 6.3 LTS and Universal Render Pipeline (URP). Hardware minimums have not been benchmarked and should not be invented.

1.5 Development status

Area	Status	Evidence / caveat
Locker room exploration	Implemented	Starter Assets third-person character and camera
Stadium javelin loop	Implemented	Run-up, aim, release, projectile flight, landing and score
Scene transition	Implemented	E interaction loads stadium scene
Cinemachine and lighting	Implemented	Three behaviors each, described in §6

1.6 Scope and monetization

This is a course prototype, not a released commercial game. No monetization, microtransactions or online economy are implemented. The intended experience is a compact single-player session, with replayability based on improving release quality and distance.

2. Player experience, objectives and progression

2.1 Player role and emotional arc

The player assumes the role of a javelin athlete. The locker room establishes anticipation and physical preparation; crossing into the stadium marks the transition from exploration to focused competition. During the run-up, rhythmic input produces urgency; the release requires precision; the flight camera creates suspense; the landing result provides immediate feedback and encourages another attempt.

2.2 Core gameplay loop

Explore preparation room → optionally open locker / collect equipment / replenish stamina / change lighting → activate stadium entrance → accelerate using Space taps → hold Left Shift to prepare → adjust angle with W/S → release Left Shift → observe Cinemachine javelin tracking → read result → press Enter for next attempt (up to three attempts).

2.3 Objectives and outcomes

Objective	Mechanic / feedback
Primary	Achieve the greatest legal javelin distance
Precision	Release close to the foul line without crossing it
Execution	Build run-up speed while managing stability
Session progression	Complete a three-attempt sequence and compare results
Failure state	Crossing the foul line produces FOUL and zero distance
Timeout	Countdown reaches zero

2.4 Key moments

Moment A — entering the stadium: player-initiated transition from preparation to competition.
Moment B — release: angle, speed, and distance from the foul line converge into a single high-stakes input.
Moment C — flight and landing: Cinemachine switches to the projectile and the measured result closes the feedback loop.

2.5 Game duration and progression limits

A run comprises three throwing attempts. Actual session duration varies with locker-room exploration and player input; it has not been measured. No leveling, skill trees, currency progression or NPC questline is implemented.

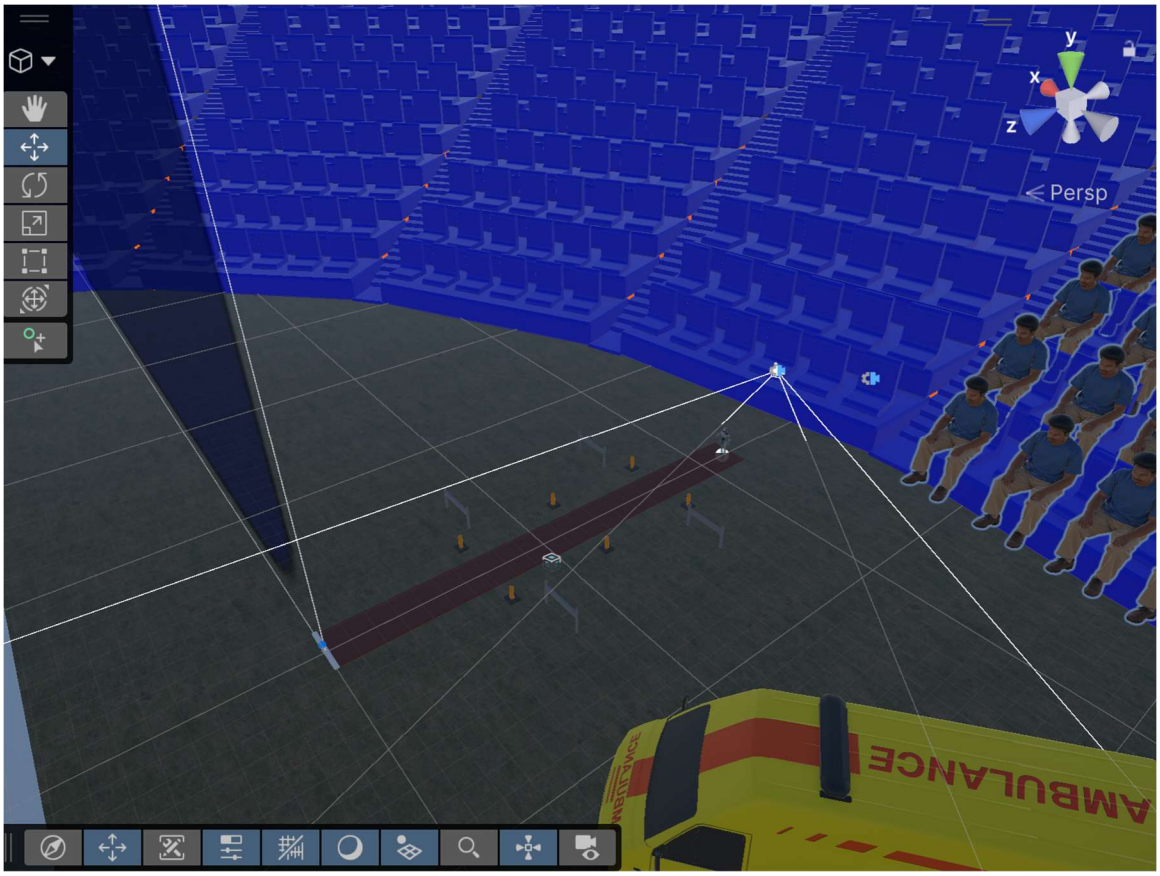
3. Game world

3.1 World organization and locations

Location	Characteristics	Player affordances
Athlete Preparation / locker room	Interior, enclosed walls, solid collision floor, locker, equipment rack, energy drink station, lighting and entrance	Walk, inspect, interact, restore stamina, transition to stadium
FinalApproach Stadium	Outdoor-style athletics setting with throwing approach, foul line, landing field, javelin physics and scoring HUD	Build speed, choose angle, throw, watch landing, retry

The two scenes are discrete spaces connected by a deliberate interaction, not a physically continuous open world. The stadium entrance presents “[E] Enter Stadium” and calls Unity SceneManager.LoadScene("FinalApproach_Stadium"). The transition is one-way in the currently confirmed prototype


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3.2 Spatial topology and navigation

The locker room supports free third-person walking within collider-bounded geometry. The stadium uses a constrained javelin approach along the forward Z direction, followed by a simulated projectile travelling into the landing field. The distinct control schemes intentionally separate preparation/exploration from competitive execution.

3.3 Physical scale and proportionality

Unity world coordinates are treated as meters for documentation and physics design. Exact dimensions of the imported locker room, runway and landing field have NOT been measured from the final Unity scene.

3.4 Boundaries and restrictions

Locker room: walls and an explicit Box Collider floor prevent the third-person CharacterController from falling through imported geometry. Stadium: the foul line is a gameplay boundary at throwLineZ = 0 in the supplied ThrowController configuration; a release beyond this line is a foul. The landing-field Collider defines valid landing evaluation. Verify stadium perimeter barriers and whether players can leave the runway.

3.5 Temporal dimension

The game advances in real time using Unity Time.deltaTime for locomotion, smoothing, light behavior and timer display. A 10-second PreparationTimer countdown is implemented as a visual timer, beginning at 10 and turning red in the last three seconds. Integration that forfeits a timed-out attempt and advances the three-attempt game state is NOT confirmed. The energy drink reappears after 15 seconds, and stamina decreases over time while moving. These time-dependent systems give the world a temporal dimension without a day/night cycle.

3.6 Environment as mechanics

The environment is not only decorative: locker doors rotate in response to E, the rack javelin disappears when collected, the energy drink restores stamina and respawns, the light switch toggles two lights, entrance proximity controls a light, and the entrance loads the competition scene.

4. World-object inventory and verification

Assignment 02 requests at least 30 different objects with role, significance and usability. The following is a 30-row audit worksheet. “Confirmed” means the object/system appeared in the development walkthrough; “Verify” means the precise instance or independent identity still needs checking in the Unity Hierarchy. This table must not be used to claim 30 verified distinct objects before completing the audit.

#	Object / distinct role	Location / usability	Status
1	PlayerArmature	Locker room / stadium — Controllable athlete visual	Confirmed
2	Player gameplay root	Stadium — Run and throw controllers	Confirmed
3	Locker	Locker room — Interactive storage fixture	Confirmed
4	Locker door	Locker room — Rotates open/closed	Confirmed
5	DoorPivot	Locker room — Door hinge transform	Confirmed
6	Javelin rack	Locker room — Equipment display	Confirmed
7	Equipment javelin	Locker room — Collectible visual disappears	Confirmed
8	Energy drink can	Locker room — Consumable stamina refill	Confirmed
9	EnergyDrinkStation	Locker room — Respawn and interaction logic	Confirmed
10	Light switch	Locker room — Manual environmental control	Confirmed
11	LockerRoomLight A	Locker room — Toggleable point light	Confirmed
12	LockerRoomLight B	Locker room — Toggleable point light	Confirmed
13	StadiumEntrance	Locker room — Scene transfer trigger	Confirmed
14	EntranceProximityLight	Locker room — Automatic distance light	Confirmed
15	LockerRoomFloorCollider	Locker room — Collision support	Confirmed
16	InteractionCanvas	Locker room — Context UI container	Confirmed
17	EnergyBar	Locker room — Stamina feedback	Confirmed
18	Locker prompt	Locker room — Open/close interaction cue	Confirmed
19	Javelin prompt	Locker room — Equipment pickup cue	Confirmed
20	Energy drink prompt	Locker room — Consumable pickup cue	Confirmed

21	Stadium player camera	Stadium — Cinemachine follow	Confirmed
22	Stadium javelin camera	Stadium — Projectile tracking	Confirmed
23	Locker close-up camera	Locker room — Interaction cinematic	Confirmed
24	PlayerFollowCamera	Locker room — Third-person exploration	Confirmed
25	Javelin projectile prefab	Stadium — Launched Rigidbody object	Confirmed
26	Landing field collider	Stadium — Landing validation	Confirmed
27	Foul line marker	Stadium — Throw validity threshold	Confirmed
28	Stadium spotlight	Stadium — Event-driven illumination	Confirmed
29	PreparationTimerText	Stadium — 10-second time display	Confirmed
30	Scoreboard / distance HUD	Stadium — Attempt result feedback	Confirmed

5. Game mechanics catalogue

Assignment 02 requires 20 distinct mechanics. The catalogue below separates verified gameplay rules from supporting behaviors; it is NOT a claim that the rubric's 20 distinct mechanics threshold has been independently met. Related controls should not be double-counted without a defensible rule distinction.

#	Mechanic	Specific rule / action	Status
1	Third-person exploration	WASD movement with CharacterController	Implemented
2	Interaction distance gating	E prompt only within proximity	Implemented
3	Locker open / close	E toggles hinged door rotation	Implemented
4	Equipment collection	E removes rack javelin	Implemented
5	Stamina depletion	Movement drains 3 units per second	Implemented
6	Stamina restoration	Energy drink refills to 100%	Implemented
7	Consumable respawn	Drink reappears after 15 seconds	Implemented
8	Manual light control	E toggles two point lights	Implemented
9	Automatic proximity light	Entrance light switches within 5 meters	Implemented
10	Scene transition	E loads stadium scene	Implemented
11	Rhythmic run-up	Repeated Space input drives forward approach	Implemented
12	Throw preparation	Hold Left Shift enters preparation state	Implemented
13	Angle adjustment	W/S adjusts throw angle 15°–60°	Implemented
14	Release timing	Shift release initiates throw	Implemented
15	Foul detection	Crossing Z=0 invalidates throw	Implemented

16	Release-quality tiers	EARLY / SAFE / GOOD / PERFECT	Implemented
17	Run-speed contribution	CurrentSpeed influences launch velocity	Implemented
18	Stability modifier	High speeds reduce multiplier	Implemented
19	Projectile ballistics	Launch velocity and Rigidbody physics	Implemented
20	Landing evaluation	Collision/event computes distance	Implemented
21	Three-attempt flow	Enter begins next attempt	Implemented
22	Camera priority transitions	Cinemachine switches based on throw event	Implemented
23	Event spotlight	Launch on, landing off	Implemented
24	Preparation countdown	10-second UI countdown	Implemented

5.1 Movement and navigation

Locker room: Starter Assets ThirdPersonController moves a CharacterController and drives third-person animation. Stadium: PlayerRunController moves the gameplay root through a rhythmic Space-tap run-up; StadiumCharacterAnimator maps root displacement to Animator parameters Speed and MotionSpeed. Root motion remains disabled to prevent double movement. Jumping, swimming, climbing and vehicle traversal are not implemented and are not needed for the javelin sport concept.

5.2 Input and interaction constraints

E interactions are proximity-dependent. Current implementation uses independent scripts listening for E, so overlapping interaction zones could trigger more than one action; stations should be spatially separated or input centralized in a future revision. The javelin pickup removes the rack model but a held hand-javelin visual is not yet confirmed.

5.3 Javelin weapon/tool semantics

The javelin is sporting equipment, not a combat weapon. It has no damage, ammo, reload, NPC targets or combat range. Its gameplay effectiveness is measured by legal landing distance.

5.4 Verified numerical rules from ThrowController

Parameter	Value / formula
Base throw velocity	18 m/s (serialized default)
Run contribution	0.9 × CurrentSpeed
Throw angle	35° initial; clamp 15°–60°
Angle adjustment speed	40°/second
Foul threshold	throwLineZ = 0 (serialized default)
Release-quality multiplier	distance > 3m: 0.70; >1.5m: 0.85; >0.5m: 0.95; otherwise 1.00
Stability multiplier	speed ≤8: 1.00; ≤9: 0.97; >9: 0.93
Launch speed	(18 + 0.9 × CurrentSpeed) × releaseQuality × stability
Forward / upward velocity	v cos(angle), v sin(angle)
Attempts	3 per session (existing gameplay)
Preparation timer	10 seconds

6. Prototype evidence and behavior

6.1 Five world-changing player actions

Action	Player input	Visible cause → effect
Open/close locker	E near locker	DoorPivot rotates; prompt updates
Collect javelin	E near rack	Equipment javelin disappears
Drink energy beverage	E near station	Energy bar fills; can disappears and respawns
Toggle room lights	E near switch	Two point lights switch state
Throw javelin	Hold/release Shift	Projectile spawns, travels and lands; result updates
Enter stadium (additional)	E near exit	Scene changes from locker room to stadium

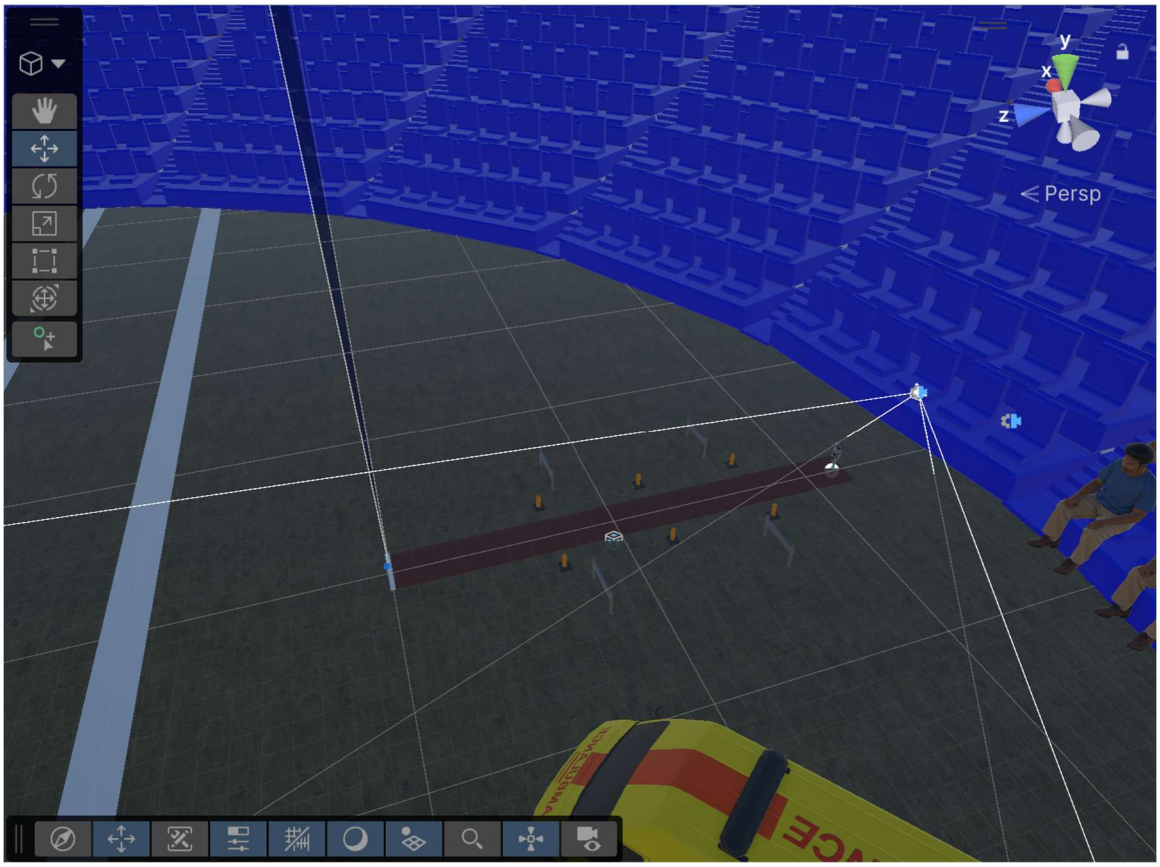
6.2 Three Cinemachine camera behaviors

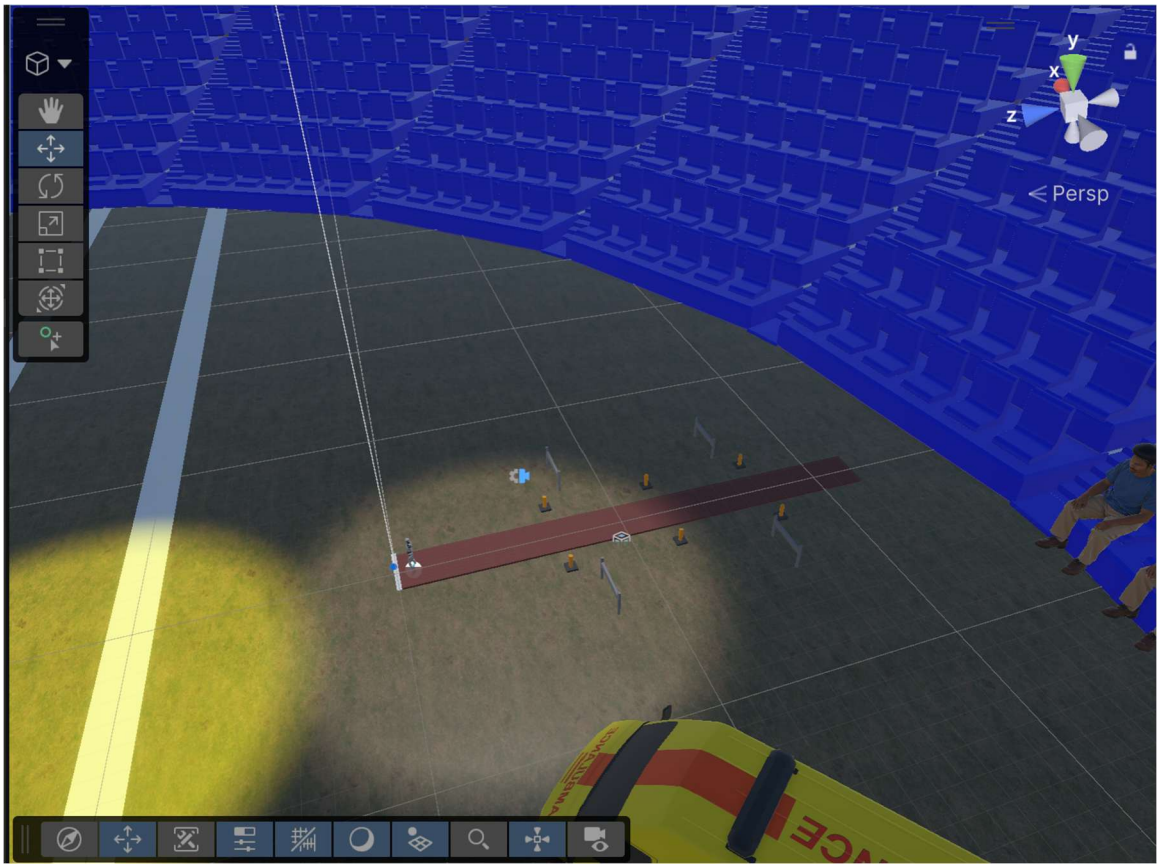
Camera	Trigger / tracking	Implementation
Preparation follow	Third-person locomotion	PlayerFollowCamera tracks PlayerCameraRoot
Locker close-up	E locker interaction	LockerInteractionCamera priority temporarily rises, then returns
Stadium javelin flight	OnJavelinLaunched	StadiumJavelinCamera tracks spawned javelin; priority swap via CameraDirector

StadiumPlayerCamera also provides pre-throw framing. The Main Camera uses Cinemachine Brain; its Default Blend duration is adjustable, with a tested short transition. The previous direct-transform CameraDirector was replaced by a priority/target-switching version.

6.3 Three distinct lighting behaviors

Lighting behavior	Trigger	Effect
Manual room lighting	Player presses E on switch	Two locker-room point lights toggle
Proximity entrance lighting	Player within 5m of entrance	EntranceProximityLight enables; disables on exit
Event-driven stadium spotlight	ThrowController launch / landed events	Spotlight enables during flight, disables on landing


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6.4 Demonstration video shot list

Recommended sequence: locker-room establishing shot → third-person walk → locker opening and close-up → javelin pickup → stamina drink and respawn → manual light switch → entrance proximity light → scene transition → run-up and angle → throw and Cinemachine follow → stadium spotlight → landing result → next attempt → timer display.

7. Interface, art, sound and accessibility

7.1 Controls

Context	Input	Action
Preparation	WASD / mouse	Third-person movement / camera
Preparation	E	Interact with nearby station / door / entrance
Stadium	Space repeated	Rhythmic run-up
Stadium	Left Shift hold/release	Prepare and launch javelin
Stadium	W / S during preparation	Increase / decrease throw angle
Stadium	Enter	Next attempt
Stadium	H	How-to-play help

7.2 HUD and interaction feedback

Preparation HUD: context-sensitive TMP interaction prompts and stamina slider. Stadium HUD: score, attempt/distance feedback, release quality, help overlay, and TIME: 10 countdown text. The countdown turns red in its last three seconds. Capture each screen and label key interface regions in the final report.

7.3 Visual and audio direction

Visual tone: readable sports venue with grounded environmental props, a recognizable third-person athlete, clear throwing field, and lighting that communicates interaction and event state. The character comes from Unity Starter Assets. The locker room and energy drink include imported 3D assets, with materials adjusted for URP. Stadium cheer/audio feedback was present in the earlier prototype. An exact audio asset inventory, license information and soundtrack plan require project inspection.

7.4 Help and usability

In-stadium H opens the existing how-to-play interface. Proximity prompts use the same E syntax to reduce cognitive load. Different movement schemes between locker room and stadium are deliberate but should be explained in the opening UI or demonstration. Color-only time warnings should ideally be paired with text or sound for accessibility.

8. Technical architecture and production

8.1 Engine and project organization

Engine: Unity 6.3 LTS, Universal Render Pipeline, C#. Core scenes: Assets/FinalApproach/Scenes/AthletePreparation.unity and Assets/FinalApproach/Scenes/FinalApproach_Stadium.unity. Unity Scene List includes both scenes. Imported character uses Starter Assets Third Person Controller; cinematic transitions use Unity.Cinemachine; text uses TextMeshPro.

8.2 Principal scripts

Script	Responsibility
PlayerRunController	Stadium run-up movement and CurrentSpeed
ThrowController	Aim, release, foul detection, launch, landing event and reset
JavelinController	Projectile physics and landing callback
CameraDirector	Cinemachine tracking targets and priorities
StadiumCharacterAnimator	Translate stadium displacement into Animator speed
LockerInteraction	Door state, prompt and close-up camera
JavelinPickup	Equipment collection
EnergyDrinkPickup / PlayerStamina	Consumable refill, respawn and stamina drain
LightSwitchInteraction / ProximityLight	Manual and automatic lighting
JavelinSpotlightController	Event-driven stadium lighting
StadiumEntrance	SceneManager scene transition
PreparationTimer	10-second countdown UI

8.3 Event and data flow

ThrowController emits OnJavelinLaunched(Transform) → CameraDirector assigns javelin tracking target and raises camera priority; JavelinSpotlightController enables spotlight. JavelinController landing callback → ThrowController emits OnJavelinLanded → CameraDirector retains landing focus and spotlight switches off; OnThrowCompleted(bool, float, string) communicates foul/legal result and release quality to scoring.

8.4 Known technical risks

Risk	Consequence	Mitigation
Independent E listeners	Two interactions could fire together	Separate trigger ranges or centralize selection
Timer timeout not scored	Player can become stuck at TIME: 0	Connect timer expiration to attempt manager
Scene-specific controllers	Character controls differ between locations	Clear HUD prompts and transition testing
Inspector overrides	Documented default values may differ	Capture Inspector values before submission
World-object count interpretation	UI/cameras may not count as distinct world objects	Verify 30 tangible scene objects
Physics repeatability	Different release positions and frame timing affect score	Run multiple reproducibility tests

8.5 Testing checklist

Test	Expected result	Record
Locker interaction	Door opens/closes, camera returns	Pass
Equipment pickup	Rack javelin disappears	Pass
Energy drink	Stamina reaches 100%; respawns in 15s	Pass
Entrance light	On at ≤5m; off farther away	Pass
Scene entrance	Loads stadium	Pass
Legal throw	Projectile flight and distance result	Pass
Foul	0m result and FOUL	Pass

Cinemachine switch	Javelin tracked and player view restored	Pass
Spotlight	On at launch; off at landing	Pass

9. Requirements and rubric crosswalk

9.1 Rubric-aligned GDD sections

The course preliminary rubric additionally values concept statement, genre, audience, player experience, key moments, art/sound, platform, competition, monetization, objectives, world, UI, MVP systems, game objects, tools, technical documentation, future ideas and prototype links. These appear throughout this document. Competition analysis and audience framing originated in Assignment 01; the latest market claims have not been re-researched for this draft.

9.2 Final-project guide versus Assignment 02

The separately uploaded final-project format guide mentions larger final deliverables such as 50 assets, five audio assets, 10 interactive elements, five NPC/enemy types and two levels. Those are not automatically Assignment 02 minimums. This document focuses on the Assignment 02 brief while preserving a structure that can expand toward the final GDD. NPC combat and enemy AI are outside the current sports prototype; do not claim they exist.

10. Detailed world walkthrough and spatial design

10.1 Athlete preparation: player journey

The AthletePreparation scene is a compact preparatory environment rather than a second competitive arena. The athlete begins with a low stamina reserve and can navigate the locker room using third-person movement. The scene teaches spatial approach, interaction range, resource recovery, equipment collection and environmental state changes before the player transitions to the competitive space. This order creates a low-risk rehearsal of interaction controls before the timing-sensitive throw.

Entry and orientation

The third-person camera tracks PlayerCameraRoot while WASD input moves PlayerArmature. The player is expected to identify interactive stations from visible props and context-sensitive TextMeshPro prompts. No minimap or objective arrow has been confirmed.

Locker interaction

Approaching the locker within approximately 2.5 Unity units exposes an E-key prompt. A pivot rotates the door toward an open angle of 90 degrees, with smooth rotation set to 180 degrees per second. The Cinemachine close-up temporarily takes priority for approximately two seconds.

Equipment collection

The javelin rack offers an E-key pickup. The displayed equipment disappears after collection. Carrying the model in the athlete's hand and enforcing equipment possession at the stadium entrance were not implemented in the confirmed build.

Energy recovery

The energy drink station restores the PlayerStamina value to 100 percent and removes the can temporarily. Its model respawns after 15 seconds. The stamina bar starts at 25 percent and decreases by 3 units per second of horizontal movement.

Lighting and departure

A manually operated light switch toggles two point lights. A separate proximity-based light activates within approximately five Unity units of the stadium entrance. Pressing E at the entrance loads FinalApproach_Stadium through SceneManager.

10.2 Stadium: competitive sequence

The stadium is a separate scene, entered through an explicit scene transition. The original Player GameObject retains PlayerRunController and ThrowController; a Starter Assets PlayerArmature child supplies the visible athlete and locomotion animation. This composition preserves the tested run-up and throw physics while improving character readability. The stadium has a runway, foul line, landing field, a score display and environmental spectator/stadium props. Exact geometry and object counts require inspection in Unity.

Stage	Player input / condition	System response	Visible feedback
Setup	Scene loaded or next attempt begins	CameraDirector activates StadiumPlayerCamera	Athlete and throwing area framed
Run-up	Repeated Space presses	PlayerRunController updates movement and speed	Athlete translates; Animator Speed parameter changes
Preparation	Hold Left Shift	ThrowController enters preparation state	W/S angle changes are accepted

Release	Release Left Shift	Launch velocity computed and javelin instantiated	Cinemachine follows flying javelin
Landing	JavelinController reports collision/landing	OnJavelinLanded and OnThrowCompleted events emitted	Distance/result and spotlight change
Next attempt	Enter through existing game flow	ThrowController.ResetThrow and camera reset expected	Athlete returns to setup state

10.3 Spatial scale measurement worksheet

The assignment asks for concrete playable dimensions. These measurements are deliberately left as verification fields because no Unity Transform bounds or scene survey has been supplied. Measure world-space extents using the Scene view grid or the coordinates of boundary objects; state the method used.

11. Object-by-object design inventory

Assignment 02 requires at least 30 different objects. The following is a 30-slot inspection register, not a claim that every named candidate has been verified as a distinct implemented object. Status “confirmed” reflects the recorded development sessions

11.01 PlayerArmature

Location: AthletePreparation. Role and usability: Controlled avatar; collision and locomotion. Evidence status: Confirmed. Interaction classification: player-operated or gameplay-driven. .

11.02 PlayerCameraRoot

Location: AthletePreparation. Role and usability: Cinemachine tracking target. Evidence status: Confirmed. Interaction classification: player-operated or gameplay-driven. .

11.03 PlayerFollowCamera

Location: AthletePreparation. Role and usability: Third-person follow behavior. Evidence status: Confirmed. Interaction classification: player-operated or gameplay-driven. .

11.04 Locker

Location: AthletePreparation. Role and usability: Interactable storage prop. Evidence status: Confirmed. Interaction classification: player-operated or gameplay-driven. .

11.05 DoorPivot

Location: AthletePreparation. Role and usability: Rotational hinge for locker door. Evidence status: Confirmed. Interaction classification: environmental, camera, UI or system object. .

11.06 Locker door mesh

Location: AthletePreparation. Role and usability: Visible door rotated by hinge. Evidence status: Confirmed. Interaction classification: player-operated or gameplay-driven. .

11.07 EquipmentJavelin

Location: AthletePreparation. Role and usability: Collectible sports implement. Evidence status: Confirmed. Interaction classification: player-operated or gameplay-driven.

11.08 JavelinRack

Location: AthletePreparation. Role and usability: Equipment placement station. Evidence status: Confirmed. Interaction classification: player-operated or gameplay-driven.

11.09 EnergyDrinkStation

Location: AthletePreparation. Role and usability: Consumable interaction station. Evidence status: Confirmed. Interaction classification: player-operated or gameplay-driven. .

11.10 Energy drink can model

Location: AthletePreparation. Role and usability: Visible collectible; respawns. Evidence status: Confirmed. Interaction classification: player-operated or gameplay-driven.

11.11 LightSwitch

Location: AthletePreparation. Role and usability: Manual lighting control. Evidence status: Confirmed. Interaction classification: player-operated or gameplay-driven.

11.12 LockerRoomLight A

Location: AthletePreparation. Role and usability: Toggleable room light. Evidence status: Confirmed. Interaction classification: player-operated or gameplay-driven.

11.13 LockerRoomLight B

Location: AthletePreparation. Role and usability: Toggleable room light. Evidence status: Confirmed. Interaction classification: player-operated or gameplay-driven.

11.14 StadiumEntrance

Location: AthletePreparation. Role and usability: Scene transition interaction. Evidence status: Confirmed. Interaction classification: player-operated or gameplay-driven.

11.15 EntranceProximityLight

Location: AthletePreparation. Role and usability: Distance-triggered light. Evidence status: Confirmed. Interaction classification: player-operated or gameplay-driven.

11.16 LockerRoomFloorCollider

Location: AthletePreparation. Role and usability: Solid floor boundary. Evidence status: Confirmed. Interaction classification: player-operated or gameplay-driven.

11.17 InteractionCanvas

Location: AthletePreparation. Role and usability: Context prompts and stamina HUD. Evidence status: Confirmed. Interaction classification: environmental, camera, UI or system object.

11.18 EnergyBar

Location: AthletePreparation. Role and usability: Resource display slider. Evidence status: Confirmed. Interaction classification: environmental, camera, UI or system object.

11.19 LockerInteractionCamera

Location: AthletePreparation. Role and usability: Close-up Cinemachine camera. Evidence status: Confirmed. Interaction classification: player-operated or gameplay-driven.

11.20 Player

Location: FinalApproach_Stadium. Role and usability: Run-up and throwing controller root. Evidence status: Confirmed. Interaction classification: player-operated or gameplay-driven.

11.21 Stadium PlayerArmature

Location: FinalApproach_Stadium. Role and usability: Animated visible athlete child. Evidence status: Confirmed. Interaction classification: player-operated or gameplay-driven.

11.22 StadiumPlayerCamera

Location: FinalApproach_Stadium. Role and usability: Pre-throw Cinemachine view. Evidence status: Confirmed. Interaction classification: player-operated or gameplay-driven.

11.23 StadiumJavelinCamera

Location: FinalApproach_Stadium. Role and usability: Flight tracking Cinemachine view. Evidence status: Confirmed. Interaction classification: player-operated or gameplay-driven.

11.24 JavelinEventSpotlight

Location: FinalApproach_Stadium. Role and usability: Launch/landing event illumination. Evidence status: Confirmed. Interaction classification: player-operated or gameplay-driven.

11.25 PreparationTimerText

Location: FinalApproach_Stadium. Role and usability: Ten-second countdown display. Evidence status: Confirmed. Interaction classification: player-operated or gameplay-driven.

11.26 PreparationTimerManager

Location: FinalApproach_Stadium. Role and usability: Countdown script host. Evidence status: Confirmed. Interaction classification: player-operated or gameplay-driven.

11.27 Javelin prefab instance

Location: FinalApproach_Stadium. Role and usability: Projectile with flight and landing physics. Evidence status: Confirmed. Interaction classification: player-operated or gameplay-driven.

11.28 Landing field collider

Location: FinalApproach_Stadium. Role and usability: Valid landing area collision reference. Evidence status: Confirmed. Interaction classification: environmental, camera, UI or system object.

11.29 Foul line / boundary

Location: FinalApproach_Stadium. Role and usability: Defines invalid overstep release. Evidence status: Verify visual object. Interaction classification: environmental, camera, UI or system object.

11.30 Scoreboard / results HUD

Location: FinalApproach_Stadium. Role and usability: Distance, attempts and outcome display. Evidence status: Confirmed. Interaction classification: environmental, camera, UI or system object.

12. Detailed mechanics specification and compliance audit

The 20-mechanic minimum is a separate assignment requirement. The catalogue below provides a clear design description of confirmed behaviors and identifies unresolved candidates. Subsystems are described independently only when they have separate inputs, rules or outcomes. Do not present a candidate as implemented until it has been tested.

12.01 Third-person locomotion — Confirmed

WASD moves the preparation athlete using Starter Assets; collision constrains travel.

Verification: demonstrate the state change in Play Mode and capture before/after evidence.

12.02 Context-sensitive interaction range — Confirmed

Interactable prompts appear only within a specified distance from the target.

Verification: demonstrate the state change in Play Mode and capture before/after evidence.

12.03 Locker door toggle — Confirmed

E changes the locker between open and closed target rotations.

Verification: demonstrate the state change in Play Mode and capture before/after evidence.

12.04 Cinematic interaction focus — Confirmed

Locker interaction temporarily raises Cinemachine close-up priority.

Verification: demonstrate the state change in Play Mode and capture before/after evidence.

12.05 Equipment pickup — Confirmed

E removes the javelin model from its rack.

Verification: demonstrate the state change in Play Mode and capture before/after evidence.

12.06 Stamina depletion — Confirmed

Horizontal movement reduces stamina by 3 units per second.

Verification: demonstrate the state change in Play Mode and capture before/after evidence.

12.07 Energy consumable recovery — Confirmed

E restores stamina to 100 percent and removes the can.

Verification: demonstrate the state change in Play Mode and capture before/after evidence.

12.08 Consumable respawn — Confirmed

The can reappears after approximately 15 seconds.

Verification: demonstrate the state change in Play Mode and capture before/after evidence.

12.09 Manual lighting switch — Confirmed

E toggles two locker-room lights.

Verification: demonstrate the state change in Play Mode and capture before/after evidence.

12.10 Proximity lighting — Confirmed

Entrance light automatically toggles within five units.

Verification: demonstrate the state change in Play Mode and capture before/after evidence.

12.11 Scene transition — Confirmed

E at StadiumEntrance loads FinalApproach_Stadium.

Verification: demonstrate the state change in Play Mode and capture before/after evidence.

12.12 Rhythmic run-up — Confirmed

Repeated Space presses drive the stadium athlete's run-up.

Verification: demonstrate the state change in Play Mode and capture before/after evidence.

12.13 Throw preparation — Confirmed

Holding Left Shift enters a throw-preparation state.

Verification: demonstrate the state change in Play Mode and capture before/after evidence.

12.14 Angle adjustment — Confirmed

W/S adjust the release angle while preparing; clamp 15–60 degrees.

Verification: demonstrate the state change in Play Mode and capture before/after evidence.

12.15 Release and projectile launch — Confirmed

Releasing Left Shift calculates and applies javelin velocity.

Verification: demonstrate the state change in Play Mode and capture before/after evidence.

12.16 Foul detection — Confirmed

A release beyond the configured throw line yields a foul.

Verification: demonstrate the state change in Play Mode and capture before/after evidence.

12.17 Release-quality calculation — Confirmed

Distance from the line maps to EARLY, SAFE, GOOD, PERFECT.

Verification: demonstrate the state change in Play Mode and capture before/after evidence.

12.18 Speed/stability modulation — Confirmed

Run-up speed contributes to launch velocity; stability reduces high-speed releases.

Verification: demonstrate the state change in Play Mode and capture before/after evidence.

12.19 Landing and scoring — Confirmed

Landing event returns distance and quality to game result handling.

Verification: demonstrate the state change in Play Mode and capture before/after evidence.

12.20 Attempt reset — Confirmed

ResetThrow clears active javelin and resets throw state.

Verification: demonstrate the state change in Play Mode and capture before/after evidence.

12.21 Flight camera switching — Confirmed

OnJavelinLaunched activates the tracking Cinemachine camera.

Verification: demonstrate the state change in Play Mode and capture before/after evidence.

12.22 Event-driven spotlight — Confirmed

Spotlight turns on at launch and off at landing.

Verification: demonstrate the state change in Play Mode and capture before/after evidence.

12.23 Preparation countdown — Confirmed

Ten-second UI countdown is confirmed

13. Physics and timing: precise rules

13.1 Release model

The supplied ThrowController defines baseThrowVelocity = 18, runContribution = 0.9, a default release angle of 35 degrees, a minimum of 15 degrees, and a maximum of 60 degrees. While preparing, W/S adjust the angle at 40 degrees per second. These are code-level values, not empirically measured real-world javelin speeds.

Distance to line = throwLineZ − player world-space Z. If this value is negative, the release is a foul and returns a failed result with distance zero. Otherwise, release quality is 0.70 when distance exceeds 3, 0.85 when above 1.5, 0.95 when above 0.5, and 1.00 at or within 0.5 units before the line.

Stability is 1.00 for player speed up to 8, 0.97 for speed greater than 8 and up to 9, and 0.93 above 9. Launch speed is (18 + playerSpeed × 0.9) × releaseQuality × stability. The projectile velocity is (0, launchSpeed × sin(angle), launchSpeed × cos(angle)). The launch is therefore directed along the positive Z axis with upward Y velocity.

13.2 Worked examples

Scenario	Distance to line	Speed	Angle	Quality	Stability	Launch speed (approx.)
Early release	4.0	7.0	35°	0.70	1.00	17.01
Good release	1.0	8.5	35°	0.95	0.97	23.64
Perfect near line	0.3	9.5	40°	1.00	0.93	24.69
Overstep foul	−0.2	9.0	35°	N/A	N/A	No projectile

These calculations illustrate the supplied formula and are not recorded playtest scores. Actual landing distance depends on JavelinController behavior, gravity, collisions and the landing-field validation.

13.3 Temporal systems and state ownership

PreparationTimer starts at 10 seconds, decreases by Time.deltaTime, turns the label red in the final three seconds, and emits OnTimeExpired at zero. The user confirmed the visual countdown.

The energy-drink respawn interval is 15 seconds and the locker camera close-up is approximately two seconds. These are temporal mechanics in addition to the ten-second preparation display. Event order matters: camera and spotlight subscribe to launch/landing notifications, while the scoring system responds to OnThrowCompleted.

14. Prototype validation plan and evidence storyboard

14.1 Five action demonstrations

Clip	Action	Before state	After state
A	Open locker	Door closed; E prompt visible	Door rotated open; close-up camera active
B	Collect javelin	Javelin visible on rack	Javelin disappears
C	Drink energy can	Stamina below maximum; can visible	Stamina full; can absent
D	Toggle light switch	Room lighting initial state	Two lights toggled
E	Throw javelin	Athlete at runway; javelin ready	Projectile launched; landing scored

14.2 Cinemachine behavior evidence

Record the PlayerFollowCamera and PlayerCameraRoot in AthletePreparation; LockerInteractionCamera priority activation on E; and StadiumJavelinCamera tracking the instantiated projectile after the launch event. Capture the Main Camera Cinemachine Brain inspector and the two stadium Cinemachine cameras. The behavior distinction is follow, interaction close-up and dynamic projectile tracking—not merely three static camera objects.

14.3 Lighting evidence

Record manual locker-room toggle, entrance proximity toggle and stadium spotlight response to launch/landing. For each, capture an initial and activated frame from the same viewpoint. Avoid interpreting exposure changes from unrelated post-processing as proof of the intended Light component state.

14.4 Test cases

Test ID	Procedure	Expected result
TC01	Approach locker	Prompt appears within interaction distance
TC02	Press E at locker	Door rotates and camera briefly changes
TC03	Walk while stamina low	Energy bar decreases while moving
TC04	Consume drink	Stamina restores and can disappears
TC05	Wait 15 seconds	Can respawns
TC06	Toggle switch twice	Lights change and restore
TC07	Approach entrance	Proximity light activates
TC08	Leave entrance	Proximity light deactivates
TC09	Press E at entrance	Stadium scene loads
TC10	Tap Space	Stadium athlete moves with animation
TC11	Hold Shift and press W/S	Throw angle remains between 15° and 60°
TC12	Release Shift	Javelin launches; camera switches
TC13	Observe flight and landing	Spotlight activates then deactivates
TC14	Cross foul line before release	Throw result is FOUL